2380109 – Introduction to Molecular Biology

Schedule	Wednesday 13:40 – 16:30, G111
Instructor	Res. Dr. Hamit İzgi – Biology Department 223 – izhamit@metu.edu.tr – Office Hours: Thursday 13:30 – 15:00

This is a must course for CENG 1st year students. No other student is allowed to take it as an elective course.

All communication will be done through ODTÜclass. Make sure you check your METU emails daily and access odtuclass26f.metu.edu.tr for lecture slides and related materials.

Attendance is not obligatory but is encouraged. In class exercises and questions will help you better learn the concepts.

Use of mobile phones is not allowed but laptops may be open to follow lecture slides and take notes. Since attendance is not scored, students who cannot give their full attention to course is welcome to leave. Joining the classroom late is not prohibited, students should find a seat without making any noise or distraction.

Textbook: We will not strictly follow a textbook. Lecture slides are the main source of information.
You can also read the relevant chapters from the textbooks: 'Campbell Essential Biology by Pearson' and 'Life: The Science of Biology by Sinauer'

Exams: Midterm exam (40%), final exam (60%). Both exams will contain multiple choice type of questions.

Midterm exam date will be announced later and will cover all the topics until that date.

Final exam will cover all the material throughout the semester.

Makeup exam will be given during final exam period to those who have medical report.

Course Description

This course introduces fundamental concepts in molecular biology for engineering students. Beginning with molecular building blocks of life, the course covers the organizations of cells, the structure of genetic material and the flow of genetic information from DNA to RNA to protein. The course then introduces evolutionary concepts, mechanisms of gene regulation and explores how biological systems can be understood using system-level and computational approaches. The course aims to provide students with a foundation in modern molecular biology and introduce engineering students to the interdisciplinary field of bioinformatics.

Week	Date	Topic
1	30 Sep	First Meeting
2	7 Oct	Biological Molecules, Cell
3	14 Oct	DNA Structure
4	21 Oct	DNA Replication, Mutation, Repair
5	28 Oct	National Holiday / No lecture
6	4 Nov	Genome, Genomics
7	11 Nov	Genetic Code
8	18 Nov	Midterm Exam
9	25 Nov	Transcription, RNA Processing
10	2 Dec	Translation & Protein folding, sorting, modifications, degradation
11	9 Dec	Regulation of Gene Expression
12	16 Dec	NGS Technologies, Protein Folding
13	23 Dec	Evolution
14	30 Dec	Q/A